

Student ID:

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz no. 1

Summer Semester 2024-2025

DURATION: 25 MIN

FULL MARKS: 15

CSE 4803: Graph Theory

Figures in the right margin indicate full marks of questions with their corresponding COs and POs in parentheses.

1. “A simple graph of n vertices is connected if it has less than $\frac{(n-1)(n-2)}{2}$ edges” — 7
Justify whether this statement is true or false. (CO2)
(PO2)
2. Let G be a graph of 4 vertices, where the vertex set is $V = \{v_1, v_2, v_3, v_4\}$. 8
Deleting one vertex, such as v_1 , v_2 , v_3 , or v_4 results in the following subgraphs respectively. Find the original graph G , and explain how you found it. (CO1)
(PO1)

